3 Product and Service Concepts for New Ventures

The Product Concept and Commercial Opportunities

OBJECTIVES

- 1. Explore the opportunities in new products and processes, and explain why manufacturing is important in a free-market economy.
- 2. Discuss the concepts of high-tech, mid-tech, and low-tech products.
- 3. Examine how products may be conceived.
- 4. Describe the model for product development as it relates to the general paradigm for creativity.
- 5. Explain why "diffusion" is a critical dimension of product development.

INTRODUCTION

- Historically, entrepreneurship has been associated with new products and extraordinary inventors.
- Most students, however, think less about new *products* and more, about new *services,* and business education is preoccupied with service careers. Consequently, little attention is given to product concepts or opportunities in manufacturing.
- This lack of attention is unfortunate because even in our so-called post-*industrial economy,* it is a strong industrial sector that drives a nation forward.
- In this section, the argument is presented for manufacturing as the cornerstone for technological innovation and significant entrepreneurial activity.

Opportunities in new products and processes, and why manufacturing is important in a free-market economy.

- Most opportunities for both products and processes arise from changes that occur in society.
- The United States has not become less reliant on manufacturing; to the contrary, new product development coupled with extraordinary new technologies has fostered an era of even more rapid infusion of manufacturing. However, the number of people involved in this sector is declining as we quickly leave an age of labour-intensive manufacturing and enter an age in which machines do the work.
- Perhaps the great challenge of the 21st century is to develop technologically sophisticated systems of work, but to manufacture that technology domestically.

A MACRO VIEW-MANUFACTURING MATTERS

- To better understand our current and future opportunities, it is important to recognise how important manufacturing is to a nation's wealth and power.

- More important, it is essential to recognise that industrialised countries are not shifting out of manufacturing into services. The so-called post-industrial economy is a myth.

- Evidence for this conclusion is found in major research studies encompassing economic activity from World War II to the present day, in which contributions to gross national product by manufacturing have remained fairly constant between 47 and 53 percent of total goods and services in the United States.

- In Ghana, contributions to gross national product by manufacturing is about 15% and is seen be declining . There is a lot of room for improvement. Let us use the U.S. as a case study.

A MACRO VIEW-MANUFACTURING MATTERS

- Unfortunately, many of these studies have been misinterpreted (or ignored), and a popular image has emerged that manufacturing is in a serious decline.

- Reports in the popular press indicate that today less than 18 percent of direct employment is in manufacturing and that only 24 percent of GNP is created through goods-producing industries. The same reports compare current trends in manufacturing with agricultural trends 50 years ago in the U.S., and both are labelled shifts in the economy.

- Specifically, agriculture currently employs only 3 percent of the work force, whereas in 1946 the figure was about 14 percent, and in 1920, about 21 percent. Using the same criteria, manufacturing employed 37 percent of the work force in 1953, currently about half that number, and fewer workers will be employed in manufacturing in the future.

A MACRO VIEW-MANUFACTURING MATTERS

- The crucial point is that these pessimistic data are correct, but they measure only *direct labour* in manufacturing finished-goods products and *direct value added* in those finished goods.
- The data do not account for inputs (materials, technology, energy, transportation, and services) required "upstream" by suppliers to final-goods manufacturers or "downstream" by those who distribute and sell manufactured products.
- A similar argument can be made for agriculture, recognising that food processing, food products manufacturing, packaging, and distributing are all part of a more comprehensive picture of agriculture.

Clarifying the Role of Manufacturing

- Using popular data that report only the value of finished goods added directly by manufacturing, the U.S. manufacturing sector contributes *directly* about 24 percent of GNP annually. This share has been fairly stable, rising slightly during the past four decades with occasional recessionary periods when steel or automotive industries have suffered setbacks.

- Add to that the *direct* share of 24 percent of GNP that originates in services that are inputs to manufacturing, and the *direct* share of 4 percent GNP that originates in transportation as an input to manufacturing, and a realistic annual contribution estimated by the manufacturing sector is, 52 percent.

Clarifying the Role of Manufacturing

- Estimates of the manufacturing sector's contribution could be much higher if energy used in production and finished goods transportation were included.
- Consequently, we can be easily misled into thinking that the United States is headed toward a service economy.
- In fact, if that transformation would ever occur, the United States could lose its status as a major economic power. Specifically, if American manufacturing dissolved (or moved "offshore"), those services, energy resources, and transportation systems currently separated as “non-manufacturing" would also dissolve (or move offshore).

Clarifying the Role of Manufacturing

- Consider what might happen if General Motors no longer produced cars domestically but instead, moved all manufacturing to Taiwan or Korea:
  - Jobs would go overseas,
  - energy used in production would be supplied overseas, most raw materials would be purchased overseas, and a majority of support services would migrate to the production location.
  - Domestic railroads and truck fleets would no longer be needed to transport raw materials, parts, tools, subassemblies, or machinery.
  - Tightly linked “downstream” enterprises such as car dealers and fleet merchandisers would become importers.
  - Loosely linked “upstream” enterprises like advertising agencies would be working for a foreign-located (and possibly foreign-owned) company. Envision how a mass exodus of manufacturing would ripple through the economy. Even the financing and insurance required for offshore or overseas production would be provided by foreign enterprises, or by other U.S. companies that have moved overseas.

Clarifying the Role of Manufacturing

- What the continuing importance of manufacturing means for the aspiring entrepreneur is that:
- first, new products will continue to be the arteries of wealth and,
- second, that suppliers of parts and services closely linked to new products will be among the best opportunities for high-growth enterprises.
- On the downside, it also means that suppliers and services closely linked to manufacturers that are moving offshore will suffer, and subcontractors who supply other manufacturers will be at the mercy of global developments by Trans-national corporations that increase their foreign interests.

The Productivity Shift

- As noted earlier, popular themes relate how America “shifted” out of agriculture and into manufacturing and now is “shifting” out of manufacturing and into post-industrial services.
- In truth, neither shift occurred.
  - In agriculture, a technological revolution made it possible not only to feed a rapidly expanding American population but also to grow surpluses for export with fewer and fewer people directly involved in agricultural jobs.
  - In manufacturing, a technological revolution is taking place with parallel implications

The Agricultural Perspective

- The productivity change in agriculture began when Cyrus McCormick introduced his mechanical reaper in 1831.
- A further change occurred when Gustavus Swift introduced systematic meat and food processing in the late 1800s. He also invented the refrigerated railcar that made rapid transportation of meat between Chicago and New York possible.
- As World War I came to a close, Campbell Soup, Heinz, Kellogg's, Kraft, General Mills, and Pillsbury, among many others, emerged as fast-growing companies with new technologies for harvesting, processing, packaging, and delivering foods.

The Agricultural Perspective

- Many entrepreneurs emerged during the depression of the 1930s and prospered through innovative methods of growing grains and vegetables. Few of these entrepreneurs have gained the same recognition as modern folk heroes, but their contributions have been impressive.
  - ✓ For example, J. R. Simplot, a potato farmer at the beginning of the Great Depression is today a billionaire who heads a multifaceted company based largely on farming and food processing. Simplot devised a way to improve potato yields by 250 percent, then developed the technology to process potatoes, reducing costs by 50 percent. By World War II, he had found a way to quick-freeze french fries, and subsequently became the major supplier of pre-packaged french fries to both the U. S. Armed Forces and McDonald's Corporation.

- Another change occurred during the early 1950s, when 40 new strains of grain were developed, and the agronomy research industry emerged to increase food production by nearly 300 percent over the next two decades.
  - ✓ This increase was due in no small way to Norman Borlaug, an Iowa farmer who became a leading researcher in grains. Borlaug developed several of the first strains of high-yield food grains that could withstand extremely harsh conditions, and in 1970 he was awarded the Nobel Peace Prize for introducing these products on a global scale.

The Manufacturing Perspective

- One only has to glance around to see how technology has influenced our lives.
- Televisions, VCRs, CD players, hair dryers, air-conditioning systems, microwave ovens, and automatic coffee makers using materials and technology unavailable several years ago are now commonplace.
- Microcomputers, fibre-optic telecommunications, surgical lasers, new medicines from biogenetic research, and robotic engineering systems are going through second and third generation changes, yet they are products with nomenclatures that have been in our vocabulary for barely a decade.

The Manufacturing Perspective

- Someone, or some team, initially challenged the status quo to develop each of these products, and behind each useful product are hundreds, perhaps thousands, of inventions that provided the technological foundations for new products.
- As noted in handout 2, integrated circuitry, transistors, vacuum tube technology, artificial language, binary code systems, mechanical calculators, and several other major scientific breakthroughs preceded microcomputers.
- In each instance, industrial productivity made quantum leaps through the innovation process of developing these products, and in turn, the products created greater leaps in industrial productivity.
- More importantly, these innovative products came about only through extraordinary changes in *process technology,* the means and methods for manufacturing products.
- Machines were necessary to manufacture a microchip the size of a pencil eraser, and methods for creating the silicon wafer had to be developed. In addition, the development of testing equipment, materials, handling mechanisms, assembly tools, and remarkable design instrumentation had to precede actual manufacturing.

The Manufacturing Perspective

- Individuals behind this vast array of modern technology include corporate research and development scientists, college professors, graduate engineering and science students, and an invisible cadre of “tinkerers” working in basements, garages, and home workshops.
- Corporate efforts, such as development of the transistor at Bell Laboratories, emerged after World War II, and the war itself opened opportunities for innovations.
- Edwin H. Land, a college professor and physicist, had worked on military optical equipment, then turned his attention to processes for polarised light and launched Polaroid Corporation in 1948.
- William Hewlett and David Packard started their electronics testing equipment business, Hewlett-Packard, in a garage in 1939. They developed the first cathode-ray tube (known today as the CRT), which led to hundreds of new monitoring and testing devices after the war.
- Another pre-war contributor, Chester F. Carlson, invented xerography in 1938, paving the way for Xerox Corporation to introduce commercial photo-reproduction equipment in 1959

The Manufacturing Perspective

- It is amazing that in 1986 nearly 70 percent of all known inventors, scientists, and researchers responsible for the totality of humankind's technical knowledge were still alive.
- Most are still alive today, and they are not necessarily old.
- Recall that William Gates III (Microsoft Corporation), Mitch Kapor (Lotus Development Corporation), and Steven Jobs and Stephen Woziiiak (cofounders of Apple Computer Corporation) are all still in their 30s

Implications of Productivity

- What has happened in agriculture and manufacturing is a rapid evolution in techniques of farming and production that have resulted in extraordinary productivity improvements.

- Fewer people are involved in direct labour in either economic sector today than in the past because fewer people are needed to produce equivalent results.

- These productivity gains are particularly apparent in developed Western nations where surpluses of food and consumer products exist.
- It is equally apparent that these gains have not taken place in other parts of the world such as Eastern Europe, China, and the Soviet Union and Africa, where food and consumer goods must be imported.

- Following the unprecedented changes in Eastern Europe at the beginning of 1990, it became apparent to new governments in Poland, East Germany, Rumania, and the Soviet Union (under Mikhail Gorbachev's domestic reform initiatives) that their priorities for the 1990s were to centre on agricultural reform and manufacturing development.

Implications of Productivity

- The implications of an evolution in productivity in free-market economies, as well as the revolution in productivity needed in socialist-based economies, centre on entrepreneurial opportunities.
- Examples cited earlier and arguments presented earlier in this Handout testify to contributions made by entrepreneurs to technological development.
- Their efforts transformed our agrarian Western economies into industrial states, and more recently, their efforts have begun to transform the nature of our industrial states from labour-intensive manufacturing to technology-driven economies.
- Without a doubt, these transformations will occur in Eastern bloc countries that have begun to reassess their priorities, both domestically in terms of productivity and globally in terms of economic trade

Implications of Productivity

- As the 20th century closes, there are extraordinary opportunities for new products, processes, technologies, and services needed in Eastern Europe, Pacific Rim countries, and developing nations in Latin America, Africa, and the Middle East.
- The opportunities are limited only by one's imagination. For example,
  - with an awakening free press in Rumania, new means of producing and distributing magazines and newspapers are needed, and
  - in East Germany, the unshackling of the media means a proliferation of television programs, stations, television sets, and transmission capabilities.

Implications of Productivity

- Although these are exceptional opportunities born of extraordinary events, there are equally exciting opportunities at home.
- Agriculture will continue to be a challenge for new products and new technologies in food-growing processes as long as population continues to expand.
- Water has become a critical resource, and there is a pressing need for new farming techniques, water control processes, and urban water technology to serve expanding cities.
- Housing systems and technologies are archaic, and a revolution in construction is needed with new materials and home-building processes emerging.
- And, of course, information technology is only beginning to find its way into consumer products, homes, and industrial production techniques.

REVIEW

- Discuss how manufacturing matters, and discuss the opportunities for new products and technologies as the world moves toward the 21st century.
- Explain how agriculture and manufacturing both continue to be vital economic sectors despite downtrends in direct employment.

PRODUCTS AND TECHNOLOGY

- *The concepts of high-tech, mid-tech, and low-tech products.*
- All three concepts are perceptual definitions dependent on the current state of technology, education, and diffusion of information.
- The electric light bulb was high tech a century ago; VCRs were considered high-tech products a decade ago; and high-powered microcomputer s are still considered high tech today.

PRODUCTS AND TECHNOLOGY

- All three technologies, however, have become commonplace, easily replicated in factories with production processes that are equally commonplace.
- Perhaps they have become ''mid tech" in the sense that they still require skilled employees to produce them, but they are no longer on the cutting edge of new technology.
- Low-tech (or no-tech) products are those requiring few skills and common materials or technologies to produce. For example, anyone with simple tools can build television stands or make plastic signs, yet low tech is very much a part of every economy. The fact that a product is simple does not mean it has no value to consumers.

PRODUCTS AND TECHNOLOGY

- At both the macro and micro levels of innovation, few products require exceptionally sophisticated scientific knowledge. Sensational new products occasionally make headlines, but most are not *high tech* in the sense of having historic significance to humankind, such as Edison's light bulb. Periodically, major inventions or discoveries emerge that are vitally important; the transistor inspired new industries and technological applications impossible with conventional electronics.

- A majority of new products, however, will evolve at the other end of the technology spectrum as *low-tech* or *no-tech* innovations. Recent examples include no-glare sunglasses, jogging shoes, automatic coffee makers, and interesting games such as Trivial Pursuit.

- Between the extremes, there are products that require moderate expertise or special resources but are themselves uncomplicated. We conveniently call these *mid-tech* products. Examples include desktop publishing software, appealing newspapers such as USA Today, ergonomic chairs, exercise machines, and keyboard synthesisers.

High-Tech Products

- A **high-tech product** is more a state of mind than a discernible entity, but calling something high-tech is useful for describing products that *currently* reflect state-of-the-art technology.
- Products we call high tech today might be considered part of our "rust-bucket" technology tomorrow. Edison was certainly engaged in high technology for his era when he fashioned a carbon-filament light bulb, but that was long ago.
- More recently, the 1984-vintage microcomputer, revolutionary at the time, has passed into a "mid-tech" stage where it can be reproduced (cloned) in back-alley Asian factories.

High-Tech Products

- The world of high technology promises breathtaking challenges for youthful entrepreneurs willing to pick up the gauntlet.
- Digital audio systems are coming onto the market. Digital technology threatens to make obsolete non-digitalised CDs, which replaced those old cassette tapes, which in turn made LP albums obsolete.
- At the same time, an enterprising group of entrepreneurs in Sunnyvale, California, has developed a laser stylus using analog signals carried by reflected light to “read” records without friction, The new laser stylus may soon make existing electromechanics used in stereos obsolete.
- Neither digital systems nor lasers were considered useful in commercial audio systems as the 1980s began; both will probably be replaced by more sophisticated systems before the next decade expires. Someone will develop these products; others will commercialise them; and still others will render them obsolete.

Mid-Tech Products

- A majority of familiar products are less sophisticated and more readily understood than high-tech innovations and they are classified as **mid-tech.**
- An insulated storm window, for instance, is easy to understand. It consists of two or more panes of glass fabricated into a framed unit and manufactured to close specifications to assure reliable insulation. The technology required to fabricate the storm window is much more complex; it requires special machinery, skilled workers, and efficient manufacturing processes.
- Many existing products have new or unusual (but far from "high-tech") applications. For example, Velcro-that familiar bristles-and-fuzz fastener-has an industrial-strength cousin. The new Velcro fastener is made with small metal hooks rather than bristles. The hooks look like hundreds of very small Js with arrowheads on a metal pad manufactured in much the same way as small integrated circuits. The new product requires a force of more than 100 pounds per square inch to pull apart, and it is strong enough to replace most ordinary metal fasteners (screws, rivets, and bolts).
- The next generation of clothes washers and dryers may be “stuck” together with strips of metal Velcro, and it will not be long before an enterprising designer creates a "Velcro-separating" tool for repair work.

Mid-Tech Products

- Mid-tech, also a state of mind assumes new uses of existing resources or methods of production that result in new products.
- This category might also include commercialisation of existing technology that evolves through proprietary knowledge. Differentiating high-tech from mid-tech products is largely a matter of perception, but unlike high-tech products that presume application of *new* knowledge, mid-tech products can often be fabricated through adaptation of existing knowledge.
- Consequently, entrepreneurs need not be inventive geniuses to pursue mid-tech innovations; they only need to be clever in developing methods to use ideas or resources in beneficial ways.

Low-Tech Products

- Perception also plays a role in defining what is meant by low tech, but we assume a rather unsophisticated viewpoint.
- Low-tech products are usually thought to be marginal changes or improvements in existing products.
  - For example, using sheets of plastic and moulds, it is not difficult to create boxes to hold floppy disks. With the same materials, most of us could fashion paper trays, desktop pencil holders, VCR tape racks, and so on. Yet someone had to pull the resources together to manufacture these products. Someone had to design the boxes, trays, and racks, and someone will improve upon them.

Low-Tech Products

- Paul Bush, president of Bush Industries, a New York firm that makes computer workstations and a successful line of wooden furniture for electronic products , was nearly bankrupt a few years ago.
- He took over a second-generation family business in wood and plastic products, began with zero cash and “negative assets” of nearly a million dollars, yet within six years had created a powerhouse company.
- Success came with the same workers, the same facilities, and the same materials that nearly broke his father's firm. The difference was that Paul Bush introduced new products with innovative designs to provide customers with useful furniture.

Low-Tech Products

- During the 1960s, cargo ships were laboriously loaded and unloaded with slow, expensive bulk-loading equipment.

- A dock worker designed the first prototype cargo container as an experiment to reduce item-by-item loading. This innovation led to improved security and rapid freight packaging. Today “containerised” shipping is the norm.

- In 1965, Tamon lwasa, a young Japanese worker tinkering with glass beads, developed the highway reflector so common today. The reflector catches and reflects headlight glare to guide motorists better at night and in foul weather conditions. During the first several years of use in Japan, the lwasa reflector reduced that nation's accident rate by more than a third.

- Other low-tech innovations include such common items as disposable razors, butane lighters, felt-tipped pens, and pop-top cans.

Low-Tech Products

- Low-tech product development still requires insight by entrepreneurs to see opportunities, and although the resulting products are often short-lived, they represent a vast array of most consumer goods.
- They are short-lived because the lack of skill or technology used in manufacturing makes it easy for new competitors to emerge, or for the existing products to be replaced with slightly improved items. Nevertheless, even in high-tech industries like computer-systems manufacturing, thousands of low-tech products are needed to enhance workstations and to provide computer users with desks, trays, diskette cases, storage bins, printer ribbons, files, and so on.
- The low-tech entrepreneur focuses on products that can be made easily, marketed quickly, and terminated with a minimum of effort. The last point is important because patents can seldom protect low-tech products, and they can be copied or replicated easy by competitors.

IDENTIFYING OPPORTUNITIES

- In a free-enterprise system, markets arise for new products and services from *wants* and *needs* of consumers.
- In each of the examples noted previously, the entrepreneurs identified opportunities based on both wants and needs.
- Paul Bush, for example, recognised that electronic products had to sit on something; home consumers wanted practical television stands, VCR cabinets, and desks designed for microcomputer workstations. However, consumers' *wanting* such products would not have been sufficient to launch a multimillion dollar business; customers also had to *need* them.
- Bush invested heavily in market research and found that existing products (such as television stands) were often flimsy and ugly. He also found that few companies made computer workstations, and that a large gap existed between functional metal tables (modified typewriter stands) and fashionable office furniture designed specifically for word processors and personal computers.
- Paul Bush recognised the opportunity and created his product line.

IDENTIFYING OPPORTUNITIES

- Yet another example is the plastic template that can be slipped over a microcomputer keyboard for quick reference to keyboard functions. Templates provide PC users with fingertip information about keys and functions for various software applications. The opportunity gap was one of useful knowledge.
- Software developers have created exciting applications from word processing to computer-aided design, but learning to use these applications usually means wading through a thick manual written by engineers. Plastic templates help make systems more “friendly,” yet they are only printed sheets of plastic costing pennies to make.

Generating Product Ideas

- Every person has tremendous experience with a wide array of products and services. We store knowledge far more complex and in quantities that far exceed the capabilities of even the most sophisticated computer system.
- Yet when students in an entrepreneurship course were asked to brainstorm new product ideas, 35 students representing nearly 735 total years of consumer experience sat numb (and mute) for the first two weeks of the course.
- With a bit of nudging, however, they formulated a rather impressive list of ideas. They were given an assignment to identify and explain the commercial viability of at least ten new products during the third week of the course. As a result, students identified nearly 800 product ideas-more than one for each year of each student's life.

Mapping New Ideas

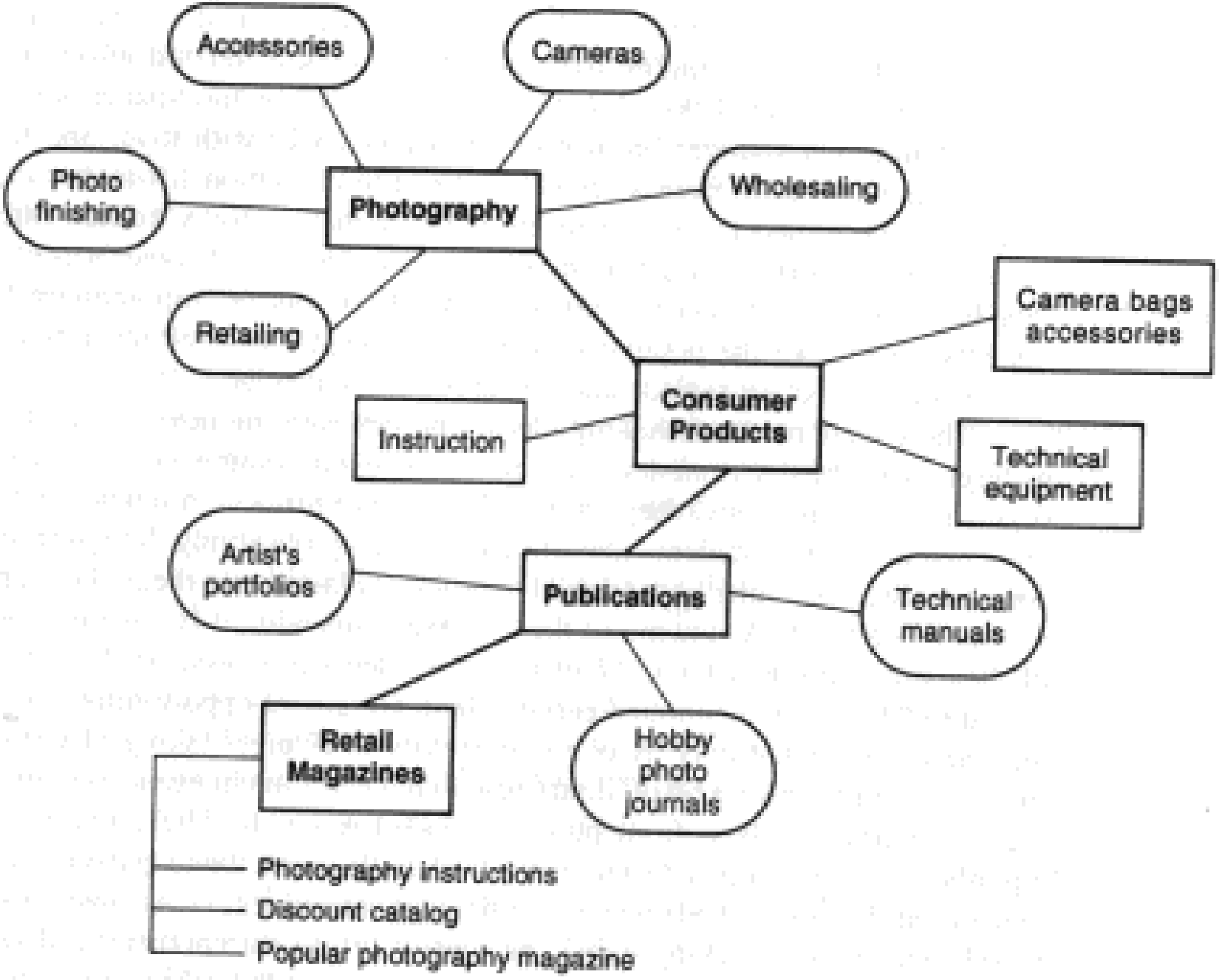
- One way to develop innovative ideas is to **mind-map.** This is a method of brainstorming in which a general idea is refined into components that represent markets, and from those components, product ideas are evaluated for their value to a particular customer. Group decision-making techniques, such as Delphi and the Nominal Group Technique (NGT), have less meaning for entrepreneurs who usually find themselves making individual decisions, yet a mind-mapping process can have interesting results.
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Mapping New Ideas

- The first stage of the mind map is to develop likely areas of product or market interests related to an entrepreneur's general business interest.
- In the illustration, the entrepreneur was an amateur photographer interested in turning a lifelong hobby into a business.
- From the first stage, she identified "consumer products- as being of more interest to her than other options. From the focus of consumer products, she extended her ideas to specific products or services, including camera bags, developing photography-instruction seminars, designing technical equipment for professional photography, and creating publications on photography.
- She selected the area of publications to explore, and in the third stage she identified several related business opportunities. Being an avid writer as well as a photographer, she quickly narrowed the field to three types of magazines that would interest her. She researched all three, and eventually decided to create a photography instruction magazine aimed at photography hobbyists and parents. (see fig 1)

Mapping New Ideas

- Although mind-mapping is simply an illustration of how one might go about identifying opportunities, it is more than a gimmick.
- It reflects the first stage in the creative process. As part of that discussion, we also described a variety of ways that opportunities arise (such as through demographic changes) and how entrepreneurs allow their ideas to "incubate- into commercially viable innovations.
- Incubation of ideas reflects the model of innovation described in detail. We build on these concepts in the next section, specifically expanding the model of a creative process as it relates to bringing a product idea to fruition.


Accessories
Cameras
Photo finishing
Photography
Wholesaling
Retailing
Camera bags accessories
Consumer Products
Instruction
Technical equipment
Artist's portfolios
Publications
Technical manuals
Retail Magazines
Hobby photo journals
Photography instructions
Discount catalog
Popular photography magazine